

```
In [1]: #pip install --upgrade seaborn
```

```
In [2]: import warnings  
warnings.filterwarnings("ignore", category=FutureWarning)
```

```
In [4]: import seaborn as sns  
sns.get_dataset_names()
```

```
Out[4]: ['anagrams',  
        'anscombe',  
        'attention',  
        'brain_networks',  
        'car_crashes',  
        'diamonds',  
        'dots',  
        'dowjones',  
        'exercise',  
        'flights',  
        'fmri',  
        'geyser',  
        'glue',  
        'healthexp',  
        'iris',  
        'mpg',  
        'penguins',  
        'planets',  
        'seaice',  
        'taxis',  
        'tips',  
        'titanic']
```

```
In [5]: len(sns.get_dataset_names() )
```

```
Out[5]: 22
```

```
In [6]: tips = sns.load_dataset("tips")  
tips.head()
```

Out[6]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

```
In [7]: tips.to_csv("tips_dataset.csv", index=False)
import pandas as pd
```

```
In [8]: import os
os.getcwd()
```

Out[8]: 'C:\\\\Users\\Suresh Kunala'

```
In [9]: import matplotlib.pyplot as plt
```

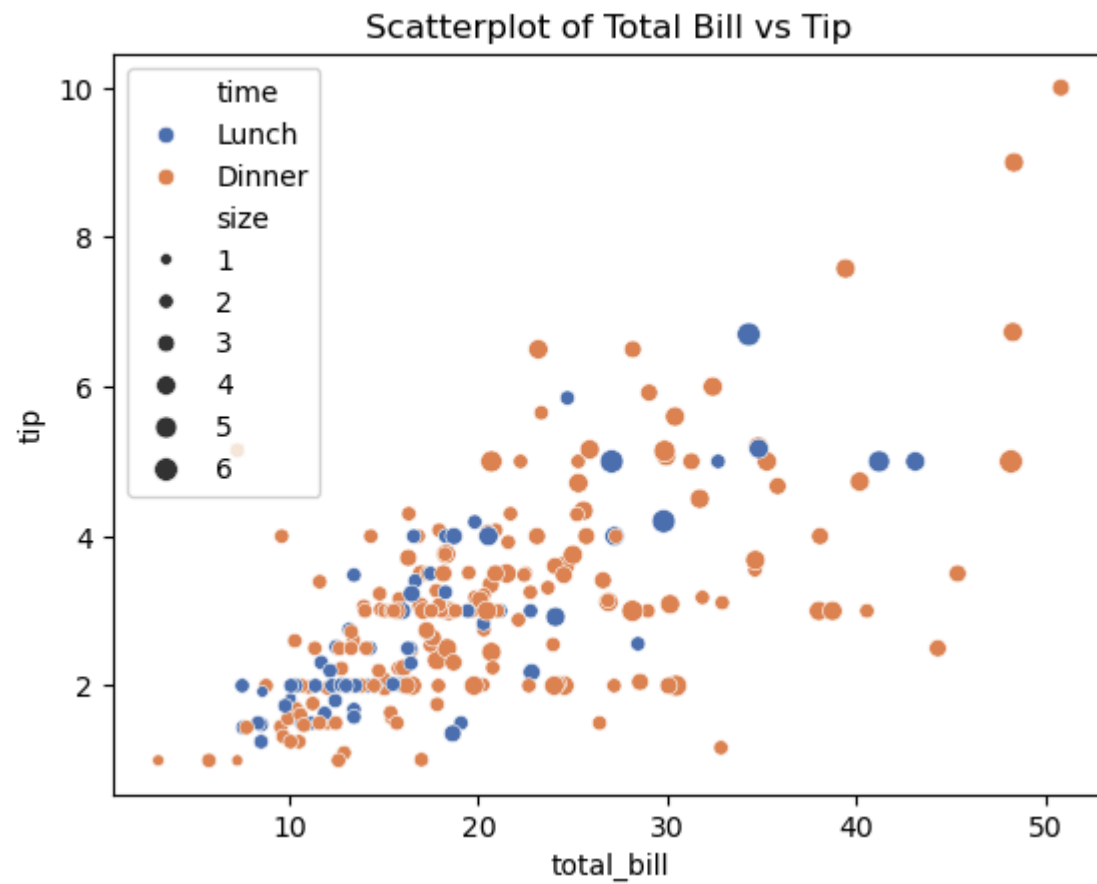
```
In [10]: plt.figure(figsize=(8, 6))
```

Out[10]: <Figure size 800x600 with 0 Axes>

<Figure size 800x600 with 0 Axes>

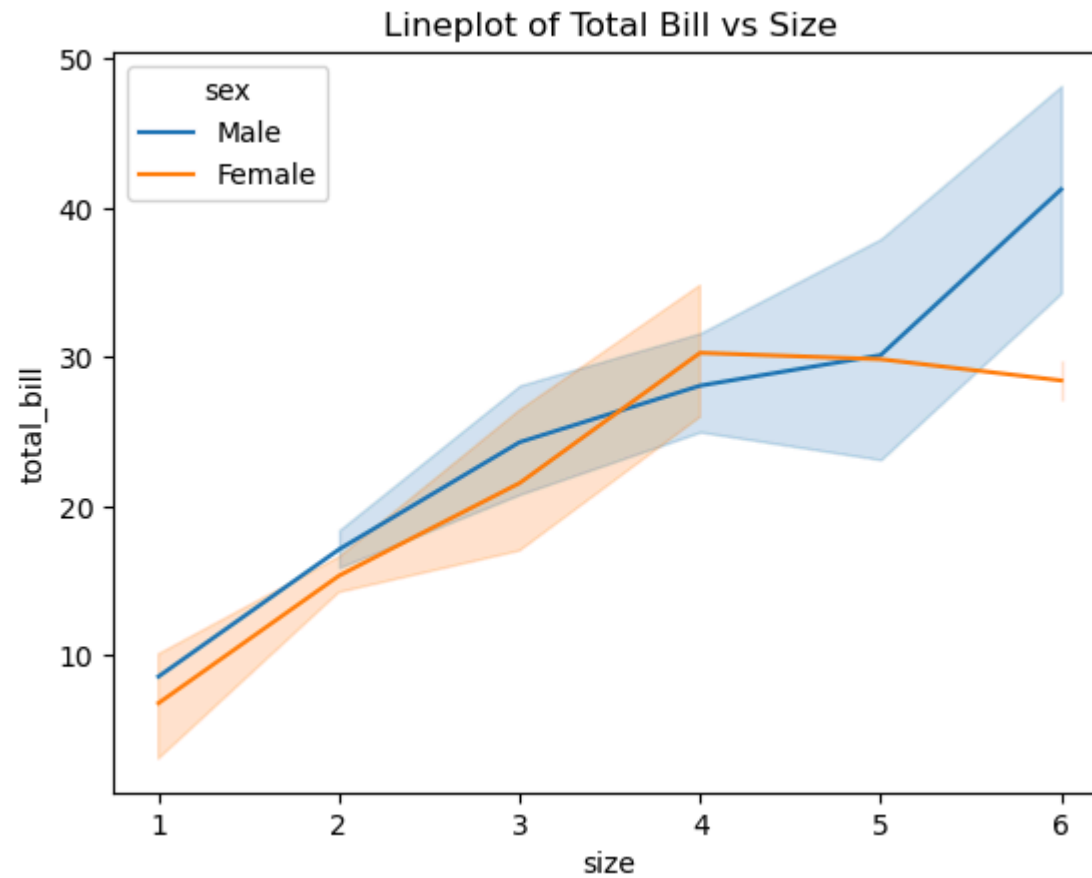
```
In [12]: # 1. scatter plot
```

```
sns.scatterplot(data=tips, x="total_bill", y="tip", hue="time", size="size", palette="deep")
plt.title("Scatterplot of Total Bill vs Tip")
plt.show()
```



In [13]: *# Line plot*

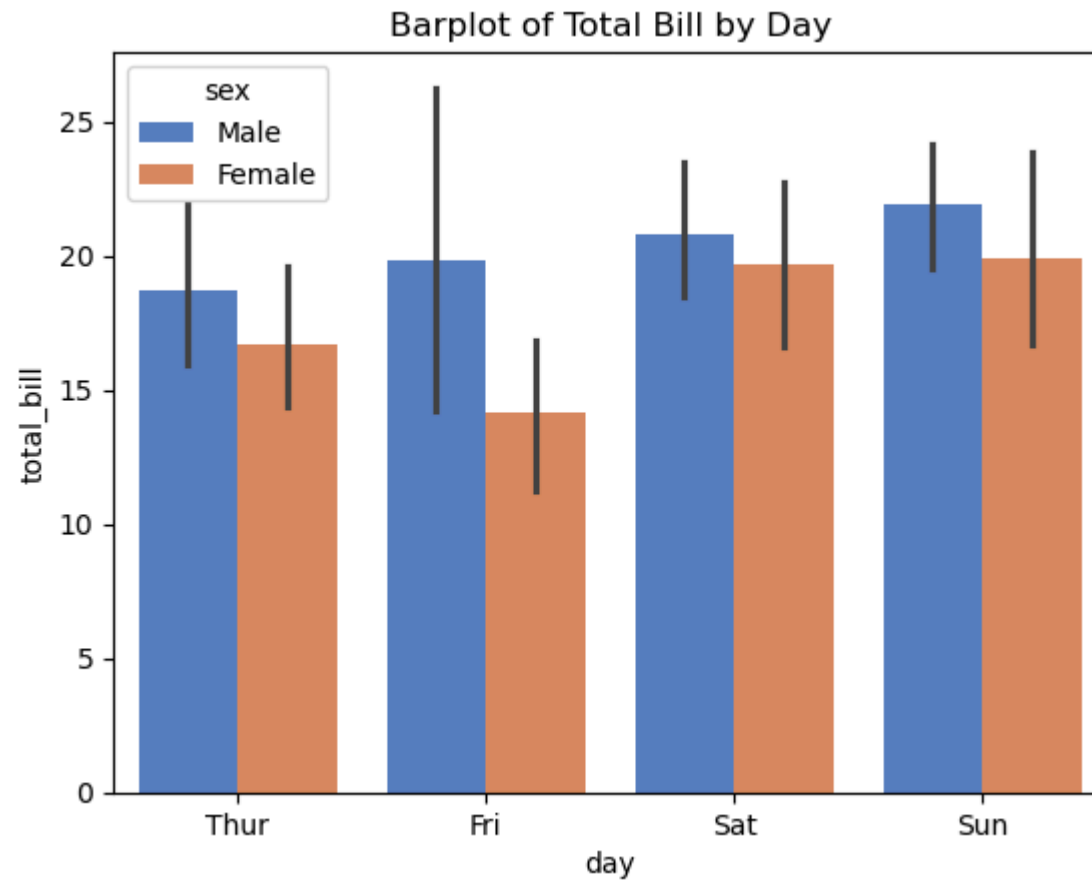
```
sns.lineplot(data=tips, x='size', y='total_bill', hue='sex', markers='o')  
plt.title("Lineplot of Total Bill vs Size")  
plt.show()
```



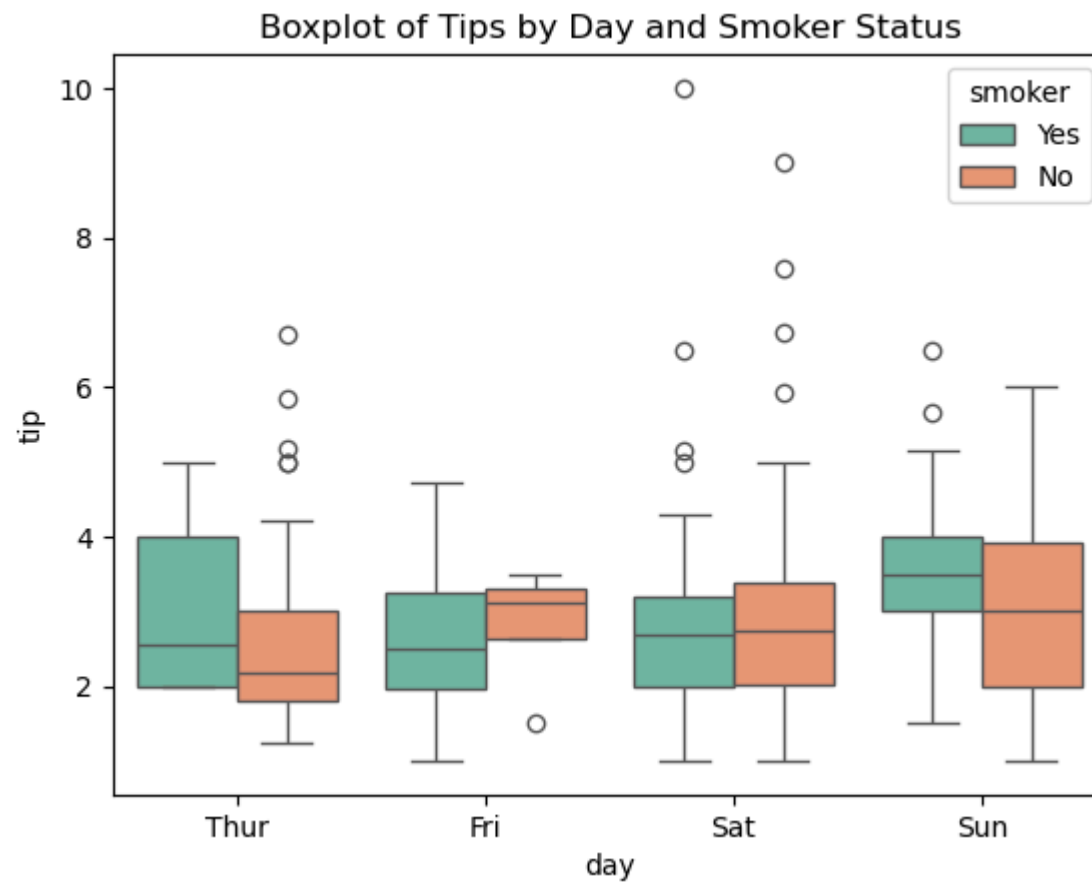
```
In [14]: tips.columns
```

```
Out[14]: Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')
```

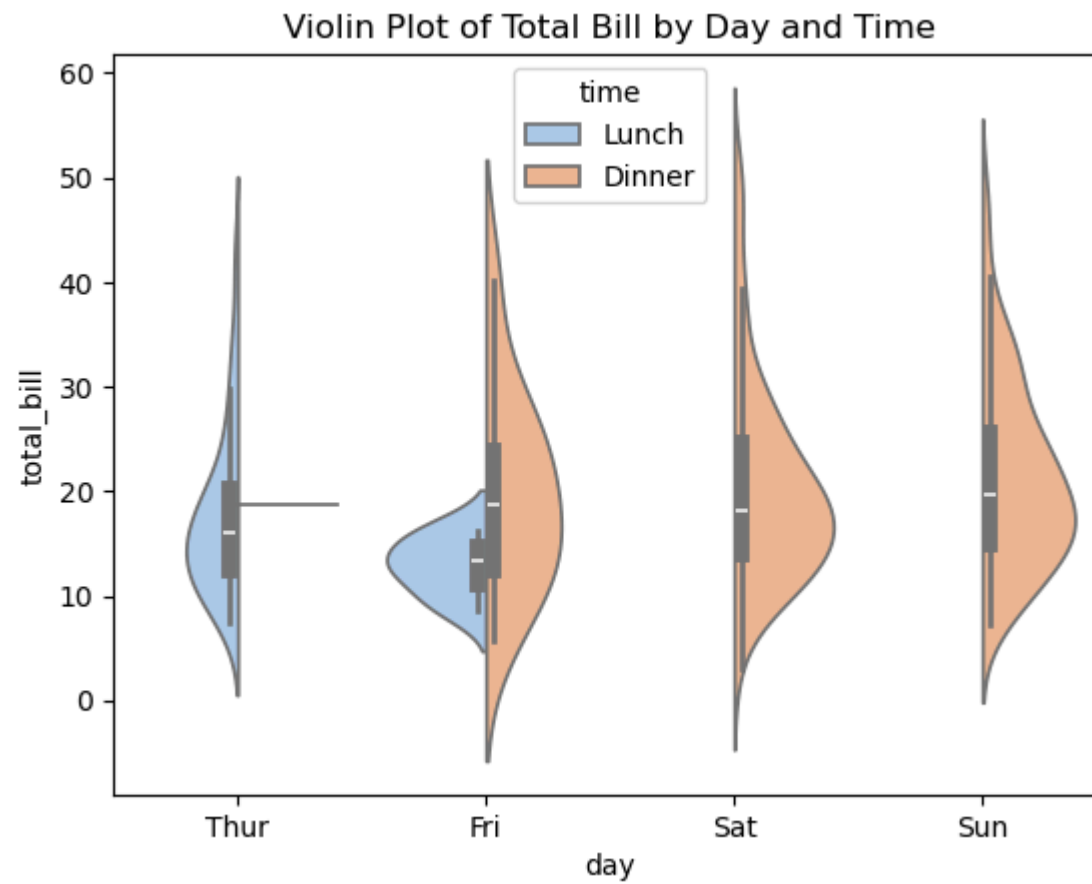
```
In [15]: sns.barplot(data=tips, x='day', y='total_bill', hue = 'sex',palette='muted')  
plt.title("Barplot of Total Bill by Day")  
plt.show()
```



```
In [16]: sns.boxplot(data=tips, x='day', y='tip', hue='smoker', palette='Set2')  
plt.title("Boxplot of Tips by Day and Smoker Status")  
plt.show()
```

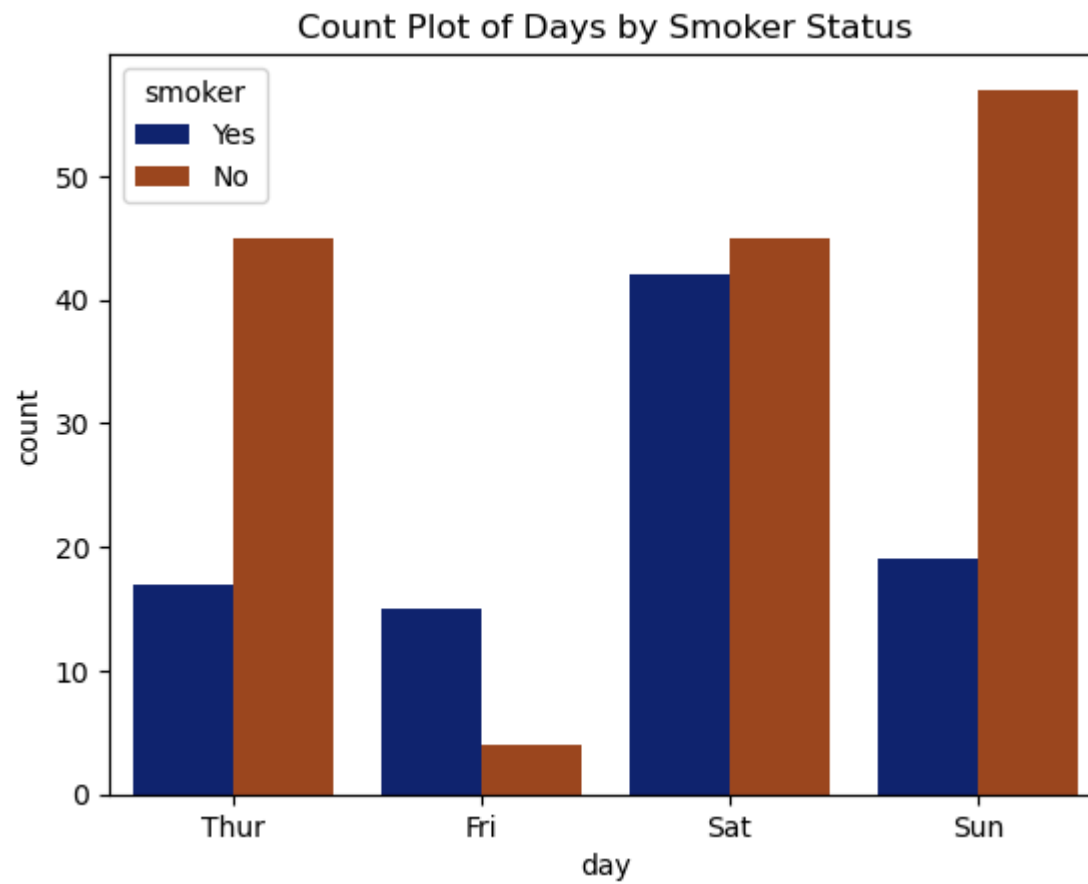


```
In [17]: sns.violinplot(data=tips, x='day', y='total_bill', hue='time', split=True, palette='pastel')
plt.title("Violin Plot of Total Bill by Day and Time")
plt.show()
```



In [18]: #6. count plot

```
sns.countplot(data=tips, x='day', hue='smoker', palette='dark')  
plt.title("Count Plot of Days by Smoker Status")  
plt.show()
```



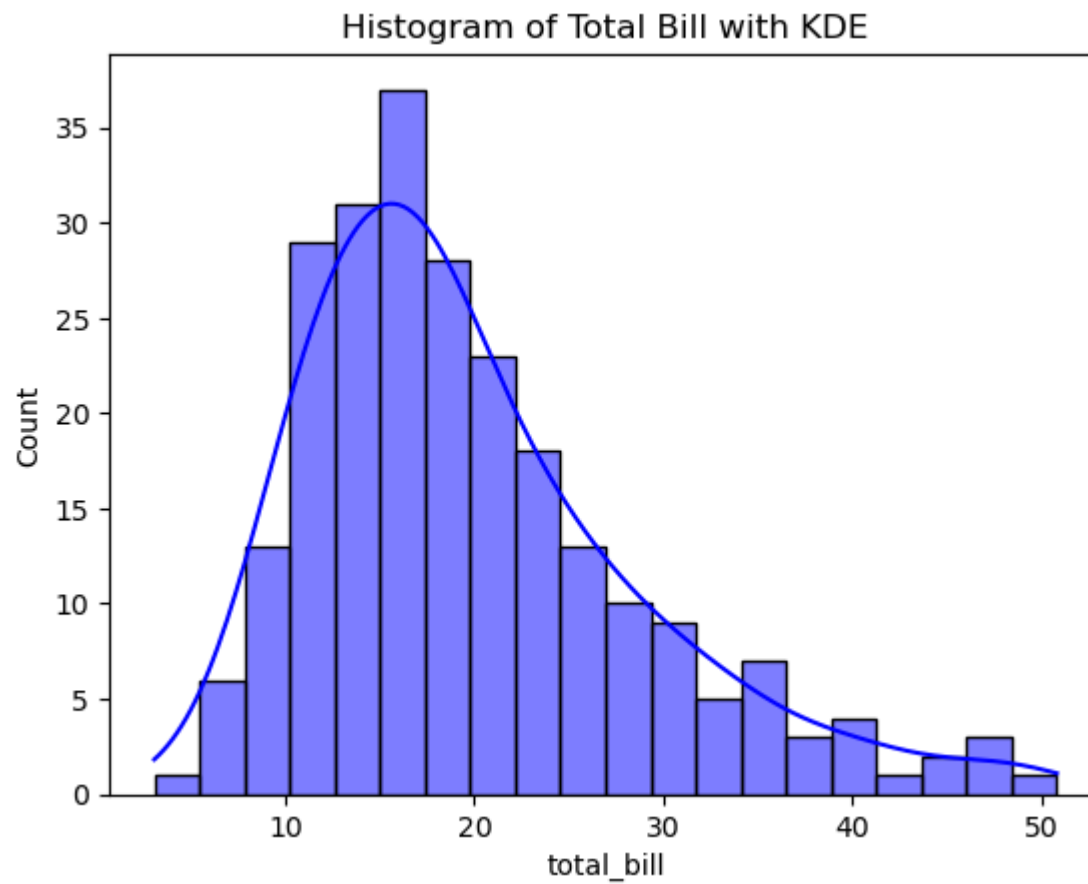
In [19]: #7. regression plot

```
sns.regplot(data=tips, x='total_bill', y='tip', scatter_kws={'s':50}, line_kws={'color':'red'})  
plt.title("Regression Plot of Total Bill vs Tip")  
plt.show()
```



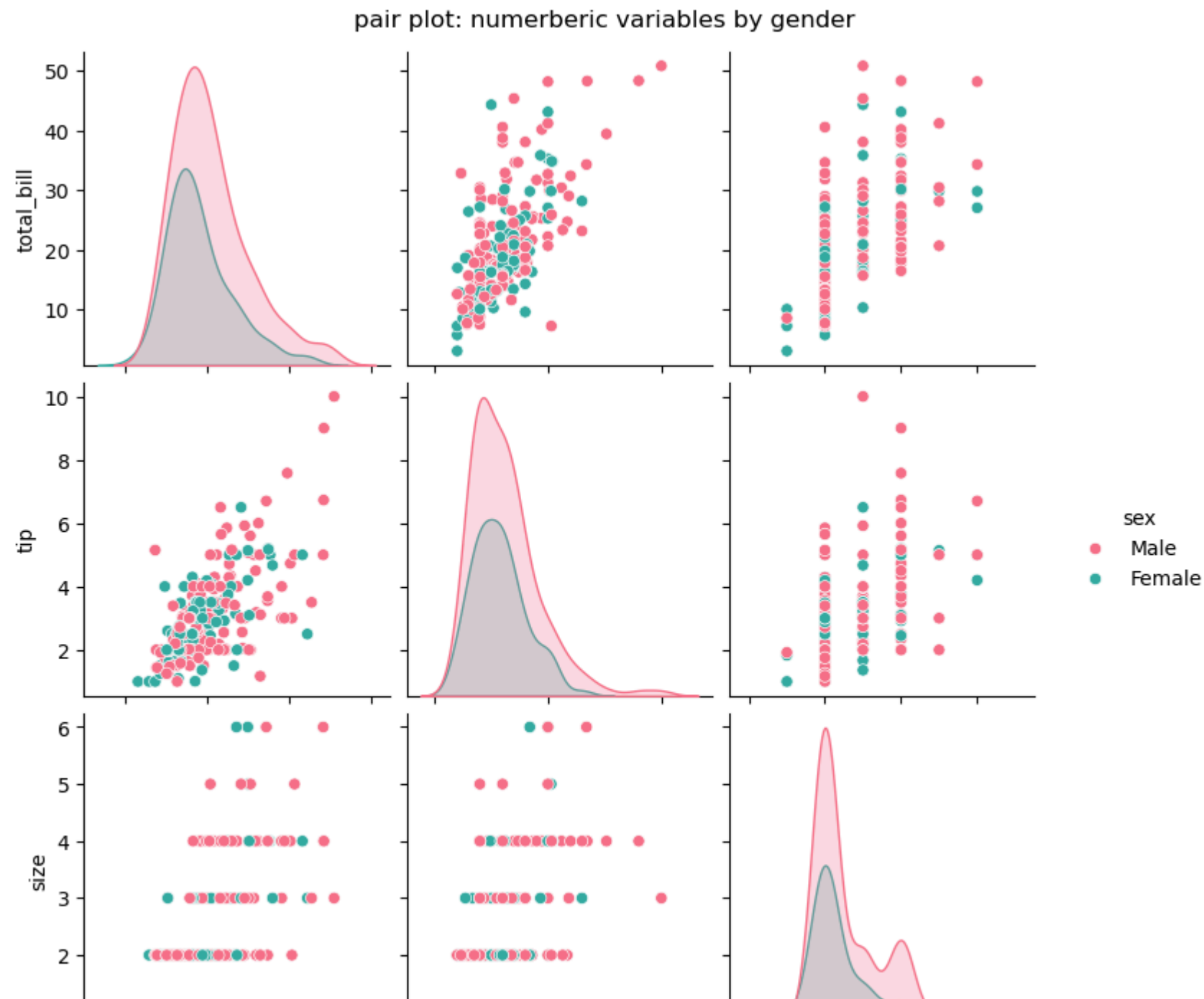

In [20]: *# 8. Histogram of total bill with KDE*

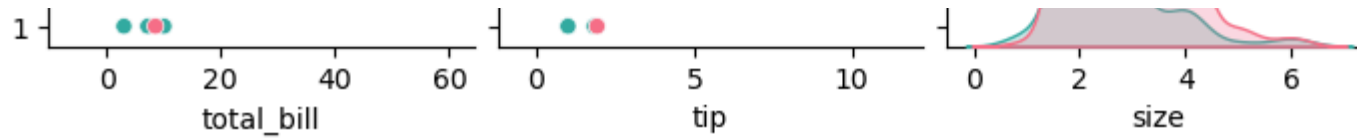
```
sns.histplot(data=tips, x='total_bill', bins=20, kde=True, color='blue')  
plt.title("Histogram of Total Bill with KDE")  
plt.show()
```



In [22]: #9. pairplot

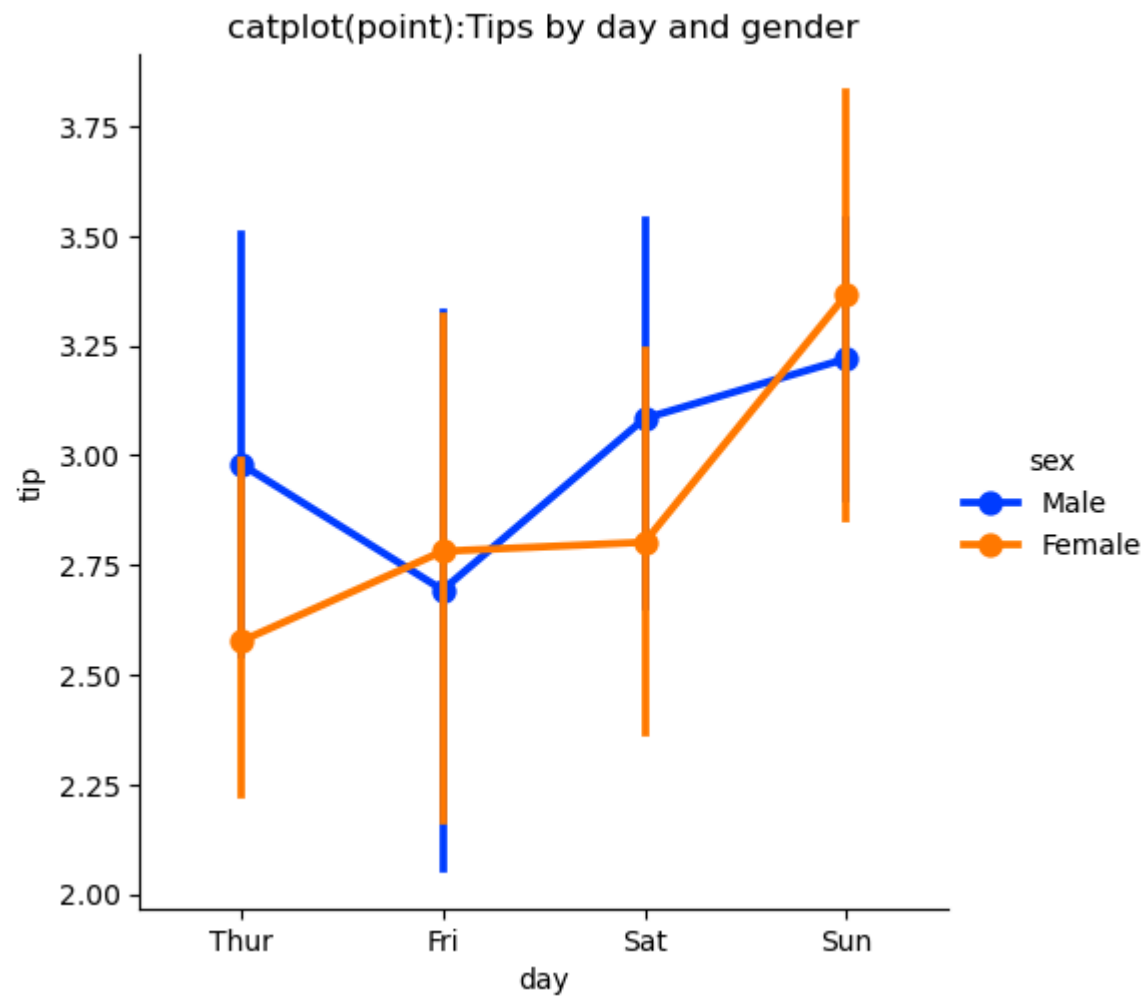
```
sns.pairplot(tips, hue='sex', vars=["total_bill", "tip", "size"], palette='husl')  
plt.suptitle("pair plot: numeric variables by gender", y=1.02)  
plt.show()
```





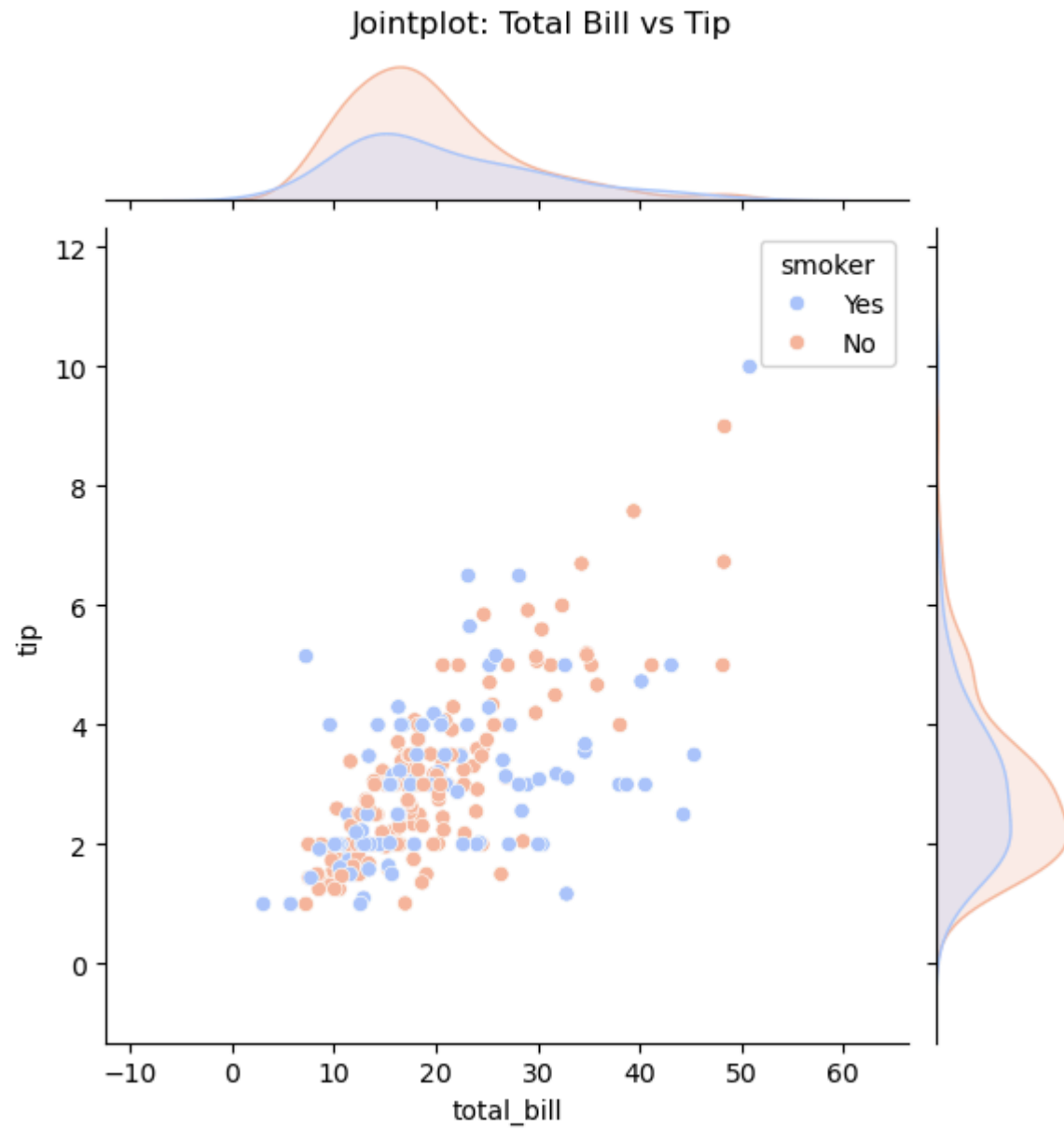
```
In [23]: # 10 catplot
```

```
sns.catplot(data=tips, x='day', y='tip', hue='sex', kind='point', palette='bright')  
plt.title("catplot(point):Tips by day and gender")  
plt.show()
```



In [24]: # 11. jointplot

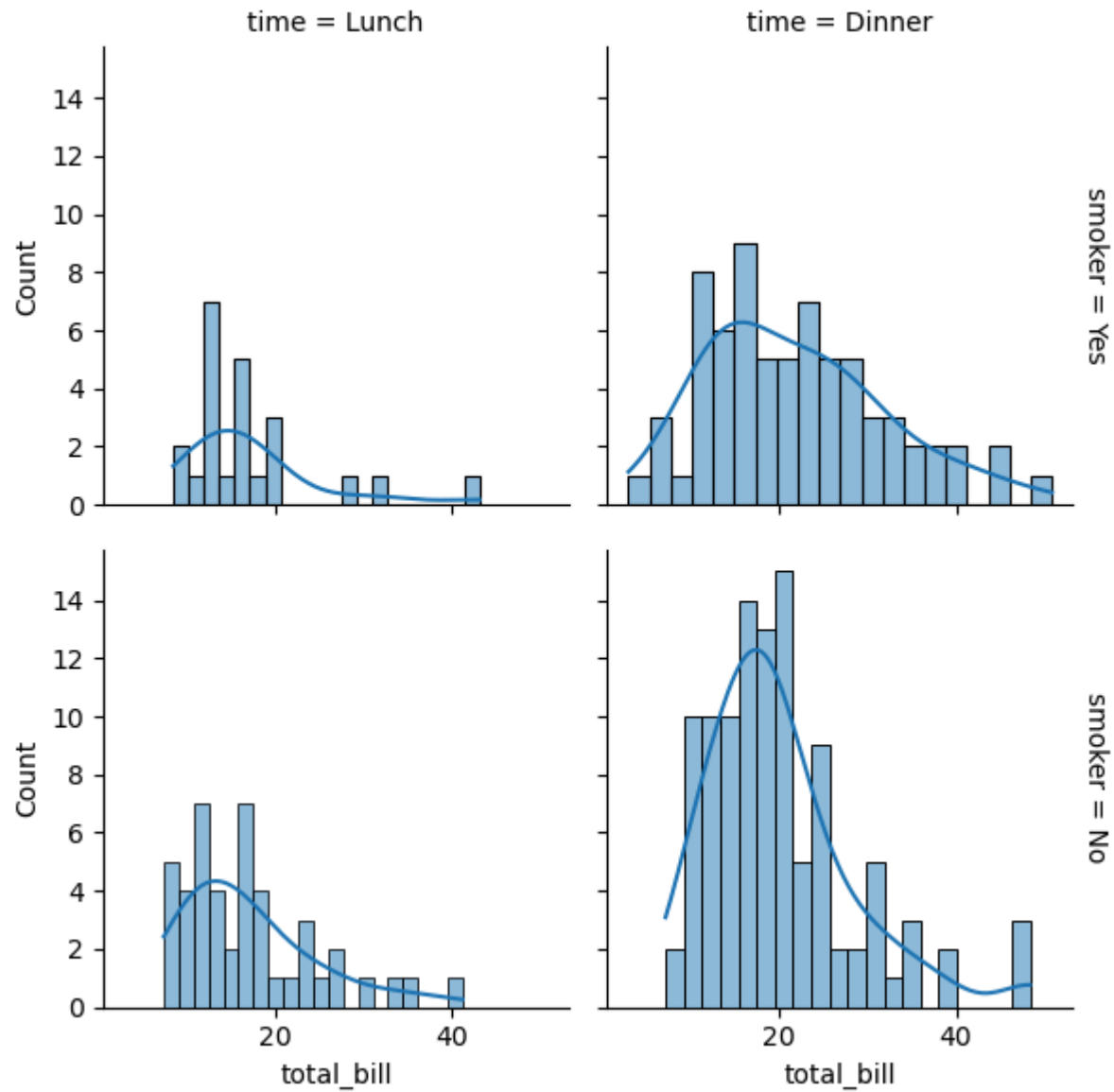
```
sns.jointplot(data=tips, x='total_bill', y='tip', kind='scatter', hue='smoker', color='purple', palette='coolwarm')  
plt.suptitle("Jointplot: Total Bill vs Tip", y=1.02)  
plt.show()
```



```
In [25]: # Facetgrid
```

```
g = sns.FacetGrid(tips, col='time', row='smoker', margin_titles=True).map(sns.histplot, 'total_bill', bins=20, kde=True).add_legend()
g
```

Out[25]: <seaborn.axisgrid.FacetGrid at 0x2ae0e451450>



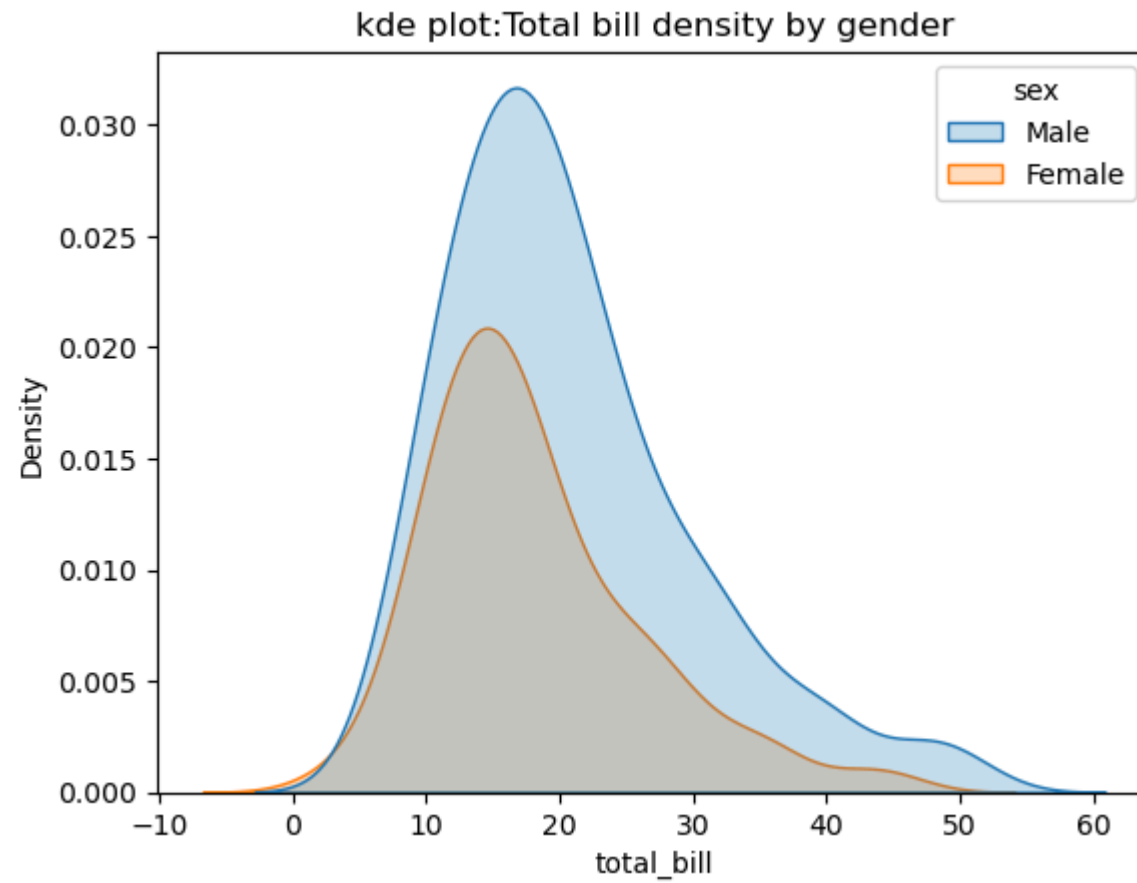
In [26]: #13. strip plot

```
sns.stripplot(data=tips, x='day', y='tip', hue='sex', jitter=True, palette='Set1')  
plt.title("strip plot: Tips by data and gender")  
plt.show()
```



In [27]: # 14. KDE PLOT

```
sns.kdeplot(data=tips, x='total_bill', hue='sex', fill=True, palette='tab10')  
plt.title("kde plot: Total bill density by gender")  
plt.show()
```

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